

Another pair of stars that is easy to spot with a small telescope is 61 Cygni (see p.252), which consists of two orange dwarfs of 5th and 6th magnitudes that orbit each other every 650 years. A large open star cluster, M39, covers an area of sky of similar size to a full moon near the constellation's border with Lacerta.

On clear nights, the Milky Way appears as a hazy band of light running through Cygnus, divided in two by an intervening cloud of dust known as the Cygnus Rift or the Northern Coalsack. The rift continues, via Aquila, into Ophiuchus.

Two large and remarkable nebulae are found in Cygnus, although neither is easy to identify.

The glowing gas cloud of the North America Nebula

(NGC 7000), near Deneb, can be glimpsed through binoculars on clear, dark nights, but its full majesty becomes apparent only on long-exposure photographs or CCD images. The Veil Nebula is a diffuse nebula found in the wing of the swan. Again, it is best seen on photographs, although the brightest part—NGC 6992—can just be made out with binoculars or a small telescope and becomes more prominent with the addition of filters to the telescope. Considerably smaller but much easier to spot is the Blinking Planetary



MYTHS AND STORIES

LEDA AND THE SWAN

The swan represents the disguise adopted by Zeus for an illicit love tryst. The object of his desire is sometimes said to have been a nymph called Nemesis or, in a more popular version, Queen Leda of Sparta. After her union with Zeus, Leda is said to have given birth to either one or two eggs, according to different versions of the story, from which hatched Castor, Pollux, and their sister Helen of Troy. Pollux and Helen were reputedly the offspring of Zeus, but Castor was the son of Leda's husband, King Tyndareus.

FAMILY GROUPING

Queen Leda, the twins Castor and Pollux, and the swan are captured in this painting after the original by Leonardo da Vinci.



OPEN CLUSTER M39

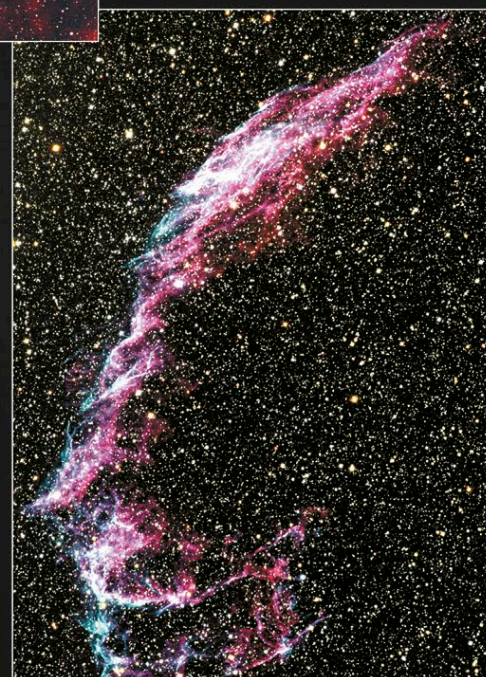
M39 is the larger and brighter of the two Messier clusters in Cygnus and contains around 30 members arranged in a triangular shape, with a double star near the center. It lies 900 light-years away and is easily spotted with binoculars. Under good conditions, M39 is visible to the naked eye.

(NGC 6826) in the other wing of the swan, with a blue-green disk similar in size to that of Jupiter. It is popularly known as the Blinking Planetary because of an odd optical effect in which, as the observer looks alternately directly at it and off to one side, it appears to blink on and off.

Two objects of considerable astrophysical interest in Cygnus are beyond the reach of amateur observers. Cygnus A (see p.324) is a powerful radio source, the result of two galaxies in collision millions of light-years away. Cygnus X-1 (see p.272), near Eta (η) Cygni, is an intense X-ray source, thought to be a black hole orbiting a 9th-magnitude blue supergiant in our galaxy.

NORTH AMERICA NEBULA

In the tail of the swan lies NGC 7000, which is popularly known as the North America Nebula, on account of its similarity in shape to that continent.



VEIL NEBULA

Splashed across an area wider than six full moons is the Veil Nebula, a loop of gas that is the remains of a star that exploded as a supernova thousands of years ago.